NanoSafe Analyzer — Research & Cytotoxicity Report

In Vitro Evaluation of ZnO Nanoparticle Biocompatibility | ISO 10993-5 Compliance Assessment

Report ID: #11 | Generated: 2026-08-17 05:08:34 | Analyst: Researcher

1. EXPERIMENT OVERVIEW

Experiment / Sample Name: Experiment test 2
Cell Line Evaluated: HeLa
Medical Application: General Biomedical
Lead Researcher: Researcher

Nanoparticle Type: ZnO
Exposure Duration: N/A
ISO 10993-5 Compliance: FAIL — Cytotoxic
Data Source: Direct Entry

2. STUDY PARTICIPANT & BIOLOGICAL SAMPLE TRACEABILITY

Biological Sample: Standard in-vitro reference cell culture line. Anonymized laboratory experiment.

3. AVERAGED CYTOTOXICITY & BIOMARKER METRICS

Mean Cell Viability: 68.91%
Reactive Oxygen Species (ROS): 3.725×
Apoptosis Inducement: 8.6375%

Mean ZnO Concentration: 49.60875 µg/mL
LDH Membrane Leakage: 12.3%

4. MACHINE LEARNING CYTOTOXICITY ASSESSMENT

Predicted Toxicity Score: 12.12 / 100
Estimated IC50 (4PL Curve): 56.2 µg/mL
ML Model Confidence: 98.5%

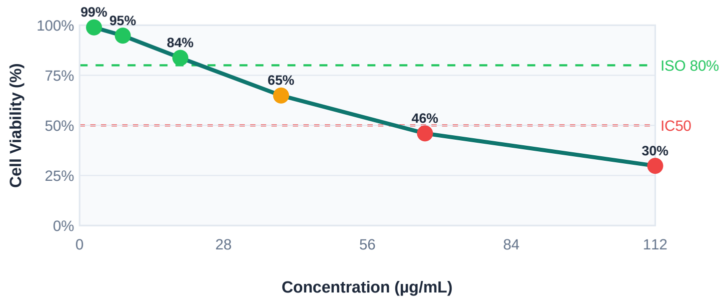
Risk Classification: Low
Biocompatible Safe Window: 0.0 - 15.7 µg/mL

5. SCIENTIFIC INTERPRETATION & RECOMMENDATION

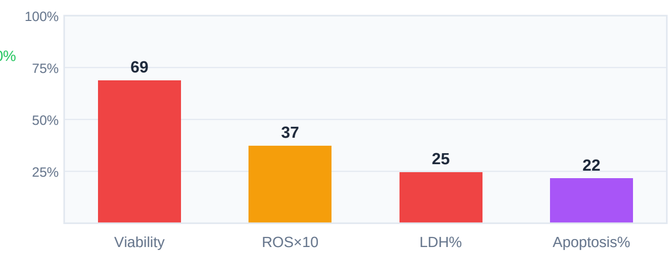
Local Scikit-Learn Ensemble Model evaluated ZnO cytotoxicity on HeLa cells under a 24.0h exposure duration for 'Wound Dressing & Topical Antimicrobial'. Measured parameters: Viability=68.91%, ROS=3.725, LDH=12.3%, Apoptosis=8.6375%. The offline ML model computed a Toxicity Score of 12.12 out of 100 (Low Risk Zone, ISO 10993-5 status: CONDITIONAL — Low-Dose Monitoring Required). Predicted IC50: 44.79 µg/mL, Safe Biomedical

6. DOSE-RESPONSE CURVE & BIOMARKER PROFILE CHARTS

Graph A: Cell Viability vs. Concentration (Dose-Response Curve)



Graph B: Multi-Biomarker Safety Profile



7. CLINICAL SUGGESTIONS & NEXT STEPS

1. **CAUTION — MODERATE TOXICITY:**
   Cell viability at 68.9% is below ISO 10993-5 PASS threshold. Formulation requires optimization before biomedical use.
2. **DOSE REDUCTION:**
   Reduce working concentration by 40-60% from current maximum. Target viability >=80% at all therapeutic dose points. Current safe ceiling: 0.0 - 15.7 µg/mL.
3. **IC50 MARGIN:**
   With IC50 at 56.2 µg/mL, reduce maximum application dose to 16.9 µg/mL (30% of IC50) to maintain adequate safety margin.
4. **SURFACE MODIFICATION:**
   Evaluate citrate or PVP surface functionalization to reduce ROS generation and membrane oxidative stress. Repeat cytotoxicity assay after coating.
5. **REPEAT STUDY:**
   Conduct 3-replicate repeat assay with optimized formulation before proceeding. Include positive control (0.1% Triton X-100) and negative control (PBS) in each plate.

Disclaimer: NanoSafe Analyzer predictive models are research aids for biomedical toxicity evaluation. Validate all findings with validated in-vitro/in-vivo protocols.